

PANCE Imaging Findings - Summary Chart

Reading approach + high-yield disease findings: CXR, KUB, CT, US (incl. FAST), Echo, MRI, and others

PANCE relevance: 'Using Diagnostic and Laboratory Studies' = 10% of the 2025 NCCPA blueprint, with skills in *selecting appropriate diagnostic studies* and *reviewing/interpreting results*. Findings below are recurring buzzwords across PA prep resources.

Chest X-Ray (CXR)

Reading Approach (RIPE technical adequacy → ABCDE-FGHI)

RIPE = Rotation (clavicles equidistant from spinous process), Inspiration (≥9-10 posterior or 5-6 anterior ribs above diaphragm), Penetration (vertebral bodies just visible behind heart), Exposure (PA vs AP - AP magnifies heart; erect vs supine).

A - Airway: trachea midline; carina; ETT tip ~4 cm above carina.

B - Bones / soft tissue: rib fractures, lytic lesions, rib notching, subQ air, breast shadows.

C - Cardiac: CT ratio <50% on PA (<60% on AP); contour; valves; pacemaker leads.

D - Diaphragm: R higher than L by ~1-2 cm; sharp costophrenic angles; **free air under diaphragm = perforation**.

E - Effusions / Everything else: pleural lines, thickening, fluid.

F - Fields (lung): symmetry, opacities, vascular markings, masses, nodules.

G - Gastric bubble under L hemidiaphragm.

H - Hila: L higher than R; lymphadenopathy; calcifications.

I - Lines / tubes / devices.

Finding	Disease	Clinical Implication / Next Step
INFECTIONS / INFILTRATES		
Lobar consolidation + air bronchograms	Bacterial pneumonia (typical, <i>S. pneumoniae</i>)	CURB-65 risk; outpatient macrolide/doxy or inpatient β-lactam + macrolide
Bulging fissure (RUL)	Klebsiella pneumonia (alcoholic, currant-jelly sputum)	Inpatient, broad-spectrum antibiotics
Bilateral diffuse interstitial / ground-glass	Atypical pneumonia (<i>Mycoplasma</i>); PCP in HIV	Macrolide for <i>Mycoplasma</i> ; TMP-SMX + steroids for PCP
Apical fibronodular opacities, upper-lobe cavitation	Reactivation tuberculosis	Airborne isolation; RIPE therapy
Calcified parenchymal nodule (Ghon focus) ± ipsilateral hilar node (Ranke complex)	Healed primary TB	Latent infection; treat per IGRA/PPD
Diffuse 1-3 mm 'millet seed' nodules	Miliary TB (hematogenous)	Same as active TB; consider HIV testing
CHRONIC / DIFFUSE LUNG DISEASE		
Bilateral hilar lymphadenopathy ± reticular markings	Sarcoidosis (Stage I/II)	ACE level, biopsy non-caseating granulomas; steroids if symptomatic
Honeycombing, basal/peripheral reticulation (HRCT)	Idiopathic pulmonary fibrosis (UIP)	Antifibrotics: pirfenidone, nintedanib
Hyperinflation, flat diaphragms, ↑ AP diameter	COPD / emphysema	Bronchodilators, smoking cessation
Eggshell hilar node calcification	Silicosis (industrial dust)	Smoking cessation; ↑ TB risk
Lower-lobe reticulonodular + pleural plaques	Asbestosis	Risk of mesothelioma, lung CA
Pleural-based mass with effusion ± plaques	Mesothelioma	Thoracoscopic biopsy
AIR / FLUID IN PLEURAL SPACE		
Visible pleural line, absent lung markings beyond	Pneumothorax	Needle decompression if tension; chest tube
Tracheal deviation AWAY from white-out	Tension pneumothorax	Immediate needle decompression (2nd ICS MCL or 4th-5th ICS AAL)
Tracheal deviation TOWARD white-out	Atelectasis / pneumonectomy / mucous plug	Bronchoscopy / re-expansion
Blunting of costophrenic angle, meniscus sign	Pleural effusion (>200 mL on PA)	Thoracentesis if symptomatic; Light's criteria for exudate
CARDIAC / VASCULAR / PE		

Finding	Disease	Clinical Implication / Next Step
Kerley B lines, cephalization, peribronchial cuffing, bat-wing/butterfly opacity, cardiomegaly	CHF / pulmonary edema	Diuresis; echo to characterize HF _r EF vs HF _p EF
Widened mediastinum (>8 cm AP at aortic knob)	Aortic dissection / mediastinal hematoma	Emergent CTA chest
Globular / water-bottle heart with clear lungs	Large pericardial effusion	Echo; pericardiocentesis if tamponade
Westermark sign (focal oligemia) / Hampton hump (wedge pleural-based opacity)	Pulmonary embolism (CXR insensitive)	Get CTPA or V/Q scan
Diffuse bilateral alveolar infiltrates with normal heart size	ARDS (PCWP <18, P/F <300)	Lung-protective ventilation (6 mL/kg IBW)
EMERGENCIES / OTHER		
Free air under right hemidiaphragm	Pneumoperitoneum (perforation)	Surgical consult
Spiculated solid pulmonary nodule >8 mm, upper lobe	Lung cancer (adenocarcinoma if peripheral)	CT, PET, biopsy; Fleischner criteria
Cavitary lesion with central necrosis	Squamous cell lung CA, abscess, TB	Tissue diagnosis
PEDIATRIC / CONGENITAL CXR SIGNS		
Steeple sign (subglottic narrowing on AP)	Croup (parainfluenza)	Humidified air, dexamethasone, racemic epi if stridor at rest
Thumbprint sign on lateral neck	Acute epiglottitis (Hib historically; now mostly other org)	Airway control before any procedures ; do NOT examine without anesthesia/ENT
Boot-shaped heart (coeur en sabot)	Tetralogy of Fallot	Surgical repair
Egg on a string with narrow mediastinum	Transposition of the great arteries	Prostaglandin E1; surgery (arterial switch)
Snowman / figure of 8	Supracardiac TAPVR	Emergent surgical repair
Figure of 3 sign / rib notching	Coarctation of the aorta	Surgical or balloon repair

Abdominal X-Ray (KUB)

Reading Approach

Confirm view - upright present? (Free air best on upright CXR or L lateral decubitus)

Bowel gas pattern: small bowel = central, **valvulae conniventes cross full lumen**, dilated >3 cm. Large bowel = peripheral 'picture frame', **haustra do NOT cross full diameter**, dilated >6 cm (>9 cm at cecum).

Air-fluid levels on upright (>2.5 cm or stepladder = obstruction).

Free air: under diaphragm; **Rigler sign** (gas on both sides of bowel wall); **football sign** in infants.

Solid organs: liver, spleen, kidney shadows, psoas margins.

Calcifications: aorta (AAA), phleboliths, renal/ureteral stones, gallstones (~10-15% radiopaque), pancreatic, appendicolith.

Bones (vertebral compression, lytic lesions) and **foreign bodies / lines**.

Finding	Disease	Clinical Implication / Next Step
Centrally dilated loops with valvulae conniventes; stepladder air-fluid levels; absent colonic gas	Small bowel obstruction (SBO)	NPO, NGT, IVF; CT to localize transition; surgery if ischemia
Peripheral colon >6 cm with haustra; cecum >9 cm	Large bowel obstruction	CT; surgical consult
Diffuse small + large bowel dilation, no transition, post-op	Paralytic ileus	Supportive (correct K+/Mg, hold opiates)
Coffee-bean / inverted-U sign pointing to RUQ	Sigmoid volvulus (older, debilitated)	Sigmoidoscopic decompression then sigmoidectomy
Distended loop pointing to LUQ	Cecal volvulus	Surgery
Transverse colon >6 cm with thumbprinting	Toxic megacolon (UC, C. difficile)	Bowel rest, IV steroids; emergent colectomy if perforation
Free air under diaphragm; Rigler / double-wall sign; football sign in infants	Pneumoperitoneum (perforation)	Emergent surgical exploration
Air in biliary tree + SBO + ectopic gallstone (Rigler triad)	Gallstone ileus	Stone removal; address fistula
Radiopaque stone along ureter, hydroureter	Nephrolithiasis (Ca-oxalate ~80% opaque)	Non-contrast CT confirms; ≤5 mm pass spontaneously
Curvilinear calcified aortic shadow >3 cm	Abdominal aortic aneurysm	US screening; CTA if symptomatic; repair ≥5.5 cm
Calcified RLQ density (appendicolith)	Suggests appendicitis	CT with contrast
Pancreatic calcifications	Chronic pancreatitis	Stool elastase; alcohol cessation; ERCP for ductal disease

Non-Contrast Head CT

Reading Approach

Confirm **non-contrast** (so blood/calcium not confused with contrast).

Symmetry - gray-white differentiation, ventricles, sulci, basal cisterns.

Blood - look for hyperdensity 60-80 HU (epidural / subdural / SAH / IPH / IVH).

Brain - gray-white preservation, sulcal effacement, mass effect, midline shift.

Cisterns - basal/suprasellar (5-pointed star); effaced = herniation risk.

Ventricles - symmetric size; trapped temporal horn = early hydrocephalus.

Bone window - skull fracture, sinuses, mastoids.

Soft tissue / scalp - hematoma, foreign body.

Finding	Disease	Clinical Implication / Next Step
Lens-shaped (biconvex) hyperdensity, does NOT cross suture lines; usually temporal	Epidural hematoma (middle meningeal artery; lucid interval)	Emergent neurosurgical evacuation
Crescent-shaped hyperdensity, CROSSES sutures but not midline	Subdural hematoma (bridging veins; elderly, alcoholic, anticoagulated)	Evacuation if symptomatic / >10 mm thick / >5 mm midline shift
Hyperdensity in basal cisterns / Sylvian fissure / sulci	Subarachnoid hemorrhage (aneurysmal; ACom most common)	'Thunderclap headache'; CTA for aneurysm; nimodipine; coil/clip. If CT negative >6 h post-onset → LP for xanthochromia

Finding	Disease	Clinical Implication / Next Step
Round/oval hyperdensity within parenchyma	Intraparenchymal hemorrhage (HTN basal ganglia; amyloid angiopathy lobar; AVM)	BP control; reverse anticoagulation; neurosurg consult
Hyperdense MCA sign, loss of gray-white differentiation, insular ribbon sign, sulcal effacement	Acute ischemic stroke (early CT signs <6 h)	tPA if <4.5 h and no contraindications; thrombectomy if LVO <24 h
Wedge-shaped hypodensity respecting vascular territory	Established/subacute ischemic stroke	Antiplatelet, statin, etiology workup
Effaced 4th ventricle, posterior fossa hyperdensity	Cerebellar hemorrhage	Surgical evacuation if >3 cm or brainstem compression
Ring-enhancing lesion(s) (with contrast)	Brain abscess; toxoplasmosis (HIV); GBM; metastasis	MRI characterizes; biopsy or empiric therapy by context

Chest CT (CT-PA / CTA)

Finding	Disease	Clinical Implication / Next Step
Intraluminal filling defect in pulmonary artery (CT-PA)	Pulmonary embolism (gold standard)	Anticoagulation (LMWH/DOAC); thrombolysis if hemodynamically unstable
RV strain, septal flattening, RV/LV ratio >1	Massive / submassive PE	Consider thrombolysis; ICU
Intimal flap with two lumens in aorta (CTA)	Aortic dissection (Stanford A = ascending; B = descending)	Esmolol then nitroprusside; surgery for type A or complicated B
Saccular/fusiform thoracic aortic dilation >5.5 cm	Thoracic aortic aneurysm	Surgical repair (Marfan, bicuspid AV at lower threshold)
Tree-in-bud opacities, centrilobular nodules	Endobronchial spread (TB, MAC, aspiration)	Cultures; targeted therapy
Subpleural reticulation + honeycombing, basal predominance	UIP / IPF	Antifibrotics (pirfenidone, nintedanib)
Anterior mediastinal mass	'4 T's': Thymoma, Teratoma, Thyroid, Terrible lymphoma	Tissue diagnosis

Abdomen / Pelvis CT

Finding	Disease	Clinical Implication / Next Step
Dilated blind-ending appendix >6 mm, periappendiceal fat stranding, ± appendicolith	Acute appendicitis	Appendectomy; antibiotics if perforated/abscess (drain)
Sigmoid wall thickening, pericolic fat stranding, adjacent diverticula	Acute diverticulitis	Outpatient PO antibiotics (uncomplicated); IV abx ± drainage if abscess
Transition point in dilated SB; whirl/whirlpool sign	Closed-loop SBO / volvulus → ischemia risk	Emergent surgery
Bowel wall thickening, pneumatosis intestinalis, portal venous gas	Bowel ischemia	Emergent laparotomy
Aortic diameter ≥3 cm; rupture → retroperitoneal hematoma	Abdominal aortic aneurysm	Repair if ≥5.5 cm, growth >0.5 cm/6 mo, or symptomatic
Hyperdense calculus in ureter; hydroureter, hydronephrosis, perinephric stranding	Ureterolithiasis	Tamsulosin; lithotripsy / ureteroscopy if >10 mm or obstructing
Dilated CBD with stone in distal duct	Choledocholithiasis	ERCP
Pancreatic enlargement, peripancreatic fluid, fat stranding ± necrosis	Acute pancreatitis (Balthazar grade)	Aggressive IVF; NPO; address cause (gallstones / alcohol)
String sign (terminal ileum narrowing)	Crohn disease	Anti-TNF, biologics; mesalamine for mild
Hepatic hypodensity with rim enhancement	Pyogenic abscess vs metastasis	Drain abscess; oncology workup if mets

Ultrasound (US) - General Principles

Reading Approach & Echogenicity Cheat

Probes: **curvilinear** (low-frequency, deep abdominal) - FAST/biliary/AAA/renal; **phased array** (small footprint, cardiac/subxiphoid); **linear** (high-frequency, superficial) - vascular, soft tissue, lung pleura, DVT.
Anechoic (black) = simple fluid (cyst, urine, bile, blood acutely).
Hypoechoic (dark gray) = complex fluid, abscess, solid hypoechoic mass.
Hyperechoic (bright) = fat, calcification, gas - **stones cast posterior acoustic shadow**.
Orientation: probe indicator → patient's right (transverse) or head (longitudinal); cephalad = top of screen.
DVT US - cornerstone is **compressibility**; non-compressible vein = thrombus.

FAST & eFAST Exam

Indications, Probe, Sequence, Performance

Indications: blunt abdominal trauma (especially hypotensive); rapid bedside triage when CT not feasible. Helps decide OR vs CT in unstable patients. Less reliable for penetrating trauma.
Probe: **curvilinear or phased-array** for the four standard FAST views; **linear** probe for the eFAST anterior chest views (pneumothorax).
Sequence (typical): 1. Pericardial (subxiphoid) → 2. RUQ (Morison's pouch) → 3. LUQ (perisplenic) → 4. Suprapubic. Add bilateral anterior chest views = **eFAST**.
Performance: sensitivity ~80% / specificity >90% for hemoperitoneum; minimum detectable free fluid ~**200-250 mL**. Trendelenburg may improve sensitivity in RUQ.
Key limitation: a **negative FAST does NOT rule out injury**. In stable patients with concerning mechanism / exam, get CT. Serial FAST may detect evolving hemorrhage.
For pneumothorax: US sensitivity ~88-90% (better than supine CXR ~50%); specificity ~99%. **For hemothorax:** US can detect as little as ~20 mL of pleural fluid.

View	Probe Placement / Landmark	What to Look For	Pearls / Pitfalls
1. Pericardial (subxiphoid; alt. parasternal)	Probe just inferior & right of xiphoid, indicator to patient's right , beam aimed toward left shoulder ; use liver as acoustic window. Alternative: parasternal long axis.	Anechoic stripe in pericardial sac → pericardial effusion. RV diastolic collapse + RA systolic collapse + plethoric IVC → tamponade physiology.	Distinguish pericardial from pleural fluid using descending aorta as landmark (pericardial is anterior to descending aorta on PS LA). Epicardial fat pad can mimic small effusion - usually anterior only.
2. RUQ (Morison's pouch / hepatorenal)	Mid-axillary line, ~10th-11th intercostal space, indicator cephalad . Visualize liver tip, hepatorenal recess (Morison's pouch), and right hemidiaphragm/subphrenic space.	Most sensitive view for hemoperitoneum in supine patient. Anechoic stripe between liver & kidney = free fluid. Also assess above diaphragm for hemothorax (anechoic above bright diaphragm).	Sweep through entire interface; small fluid pockets often at hepatic tip or caudal to liver. Perinephric fat may mimic free fluid (perirenal fat is hypoechoic but follows kidney contour).
3. LUQ (splenorenal / perisplenic)	More posterior & cephalad than RUQ - 'knuckles to the bed', ~8th-10th intercostal space at posterior axillary line, indicator cephalad . Spleen as window.	Free fluid often appears between spleen and diaphragm (subphrenic) first - look there before splenorenal interface. Also assess above diaphragm for hemothorax.	Spleen is smaller and more posterior than liver - rib shadows commonly obscure view; reposition probe between rib spaces. Stomach gas is a confounder.
4. Suprapubic (pelvic / pouch of Douglas)	Just superior to pubic symphysis in midline; rock probe down into pelvis. Scan both transverse & longitudinal . Best with full bladder (acoustic window).	Free fluid in pouch of Douglas (rectouterine, F) or rectovesical pouch (M) . In females also evaluate for IUP vs ectopic in setting of pregnancy + trauma.	Empty bladder = poor view; consider clamping Foley if needed. Pelvic free fluid is normal in small amounts in reproductive-age females (physiologic). Bowel gas can mimic or obscure.
5. eFAST - Anterior chest (pneumothorax)	Linear probe, midclavicular line at 2nd-4th intercostal spaces bilaterally, indicator cephalad . Identify pleural line between two ribs ('bat sign').	Lung sliding (pleural line shimmers with respiration) = NO pneumothorax . Absent sliding + 'barcode/stratosphere sign' on M-mode = pneumothorax . Lung point (transition between sliding and non-sliding) is pathognomonic.	Bullae, pleural adhesions, mainstem intubation, prior pleurodesis → false absent sliding. Sensitivity ~88-90%, specificity ~99%. Confirm with upright CXR or CT if equivocal.

Non-Trauma Ultrasound Findings

Finding	Disease	Clinical Implication / Next Step
ABDOMEN / PELVIS		
Gallstones with posterior acoustic shadowing, GB wall >3 mm, pericholecystic fluid, sonographic Murphy	Acute cholecystitis	NPO, IV antibiotics, cholecystectomy
GB wall thickening alone (no stones), critically ill	Acalculous cholecystitis	Percutaneous cholecystostomy
Dilated CBD >6 mm (>10 mm post-cholecystectomy)	Choledocholithiasis / obstruction	MRCP / ERCP
Target / donut sign (bowel-in-bowel) RLQ in toddler	Intussusception	Air or contrast enema
Non-compressible blind-ended tubular structure RLQ >6 mm	Acute appendicitis (preferred 1st imaging in children, pregnancy)	Appendectomy
Aortic diameter ≥3 cm	AAA	Surveillance vs repair (≥5.5 cm)
Dilated renal pelvis / calyces	Hydronephrosis	Identify obstructive cause
VASCULAR / SOFT TISSUE / NECK		
Non-compressible vein with acute soft thrombus	Deep vein thrombosis	Anticoagulation
Carotid stenosis >50% by Doppler velocities	Symptomatic carotid disease	CEA if 70-99% symptomatic stenosis
Solid hypoechoic thyroid nodule, microcalcifications, taller-than-wide, irregular margins	Thyroid malignancy risk (TI-RADS 4-5)	FNA per TI-RADS criteria
Hypoechoic fluid collection with internal debris, posterior acoustic enhancement	Soft-tissue abscess	I&D
'Cobblestoning' of subcutaneous tissues	Cellulitis	Antibiotics
OB / PELVIC		
IUP confirmed: intrauterine gestational sac with yolk sac (~5-6 wk); FHT by 6-7 wk; β-hCG discriminatory zone ~1500-2000	Intrauterine pregnancy	Routine OB care
Empty uterus + adnexal mass + free fluid + β-hCG above discriminatory zone	Ectopic pregnancy	Methotrexate vs surgery
'Snowstorm' uterus, no fetal parts, theca-lutein cysts, very high β-hCG	Hydatidiform mole	D&C; serial β-hCG monitoring
LUNG / CARDIAC POCUS (NON-TRAUMA)		
≥3 B-lines per field (lung US)	Pulmonary edema (cardiogenic) / interstitial process	Diuresis vs investigate

Echocardiogram (TTE)

Reading Approach

Confirm views - parasternal long-axis, short-axis (mitral / papillary / apex), apical 4/2/3/5-chamber, subxiphoid, suprasternal.

Chamber sizes - LV (LVEDD, LVESD), RV (>1:1 with LV = dilated), LA, RA.

Wall thickness - septum, posterior wall (>11 mm = LVH).

Systolic function (LVEF): HFrEF ≤40% / HFmrEF 41-49% / HFpEF ≥50%.

Regional wall motion - 17-segment model; akinesis/hypokinesis suggests prior MI in coronary distribution.

Diastolic function - E/A ratio, E/e' (>14 elevated), TR jet for PASP.

Valves - morphology, regurg jets (color Doppler), gradient/area (CW Doppler).

Pericardium, aortic root, IVC (size & collapse for volume status).

Finding	Disease	Clinical Implication / Next Step
Dilated LV with EF ≤40%, global hypokinesis	HFrEF / dilated cardiomyopathy	GDMT: ARNI/ACEI, β-blocker, MRA, SGLT2i; ICD if EF ≤35%
Preserved EF ≥50%, elevated E/e', LA enlargement, LVH	HFpEF	Diuresis, treat HTN, SGLT2i
Asymmetric septal hypertrophy, SAM of MV, dynamic LVOT obstruction	Hypertrophic cardiomyopathy	β-blocker; avoid dehydration / vasodilators; mavacamten / myectomy
Calcified AV: peak velocity ≥4 m/s, mean gradient ≥40 mmHg, AVA ≤1.0 cm²	Severe aortic stenosis	AVR (TAVR if high surgical risk)
MV thickening, hockey-stick deformity, MV area ≤1.5 cm²	Mitral stenosis (most often rheumatic)	Anticoagulate if AF; balloon valvuloplasty
Eccentric MR jet, dilated LA/LV	Mitral regurgitation	Surgery if symptomatic or EF declining
Mobile echodensity (vegetation) on valve, regurgitation, abscess	Infective endocarditis	Modified Duke criteria; IV antibiotics ± surgery
RWMA in coronary distribution, thinned scar, ± apical aneurysm/thrombus	Prior MI	Anticoagulate if thrombus; revascularization assessment
Effusion + RV diastolic collapse, RA systolic collapse, plethoric IVC	Cardiac tamponade	Pericardiocentesis
RV dilation, septal flattening, McConnell sign (apical sparing)	Acute PE with RV strain	CTPA; thrombolysis if massive
LA mass attached to fossa ovalis, ball-on-stalk	Atrial myxoma	Surgical excision
Apical ballooning post-emotional stress, EF recovers	Takotsubo (stress) cardiomyopathy	Supportive; rule out CAD

MRI - When MRI > CT

Brain: MS plaques, ischemic stroke beyond first hours (DWI), posterior fossa, tumor characterization, dementia workup. **Spine:** cord compression, cauda equina, disc herniation, spondylodiscitis. **MSK:** meniscus / ACL tears, rotator cuff, labrum, AVN of femoral head (early), stress fracture, osteomyelitis. **Avoid with:** pacemakers (most), metallic ocular foreign body.

Finding	Disease	Clinical Implication / Next Step
Periventricular ovoid T2/FLAIR lesions perpendicular to ventricles ('Dawson fingers'); lesions disseminated in time/space	Multiple sclerosis	DMT (ocrelizumab, etc.); MRI brain & spine
Restricted diffusion on DWI, dark on ADC	Acute ischemic stroke (most sensitive in first hours)	Stroke pathway
Cord T2 hyperintensity with vertebral body abnormality compressing cord	Metastatic spinal cord compression	Steroids, urgent radiation/surgery
Disc herniation effacing nerve root or thecal sac with saddle anesthesia / retention	Cauda equina syndrome	Emergent decompression
Crescent-shaped subchondral T1 hypointensity in femoral head	Avascular necrosis (steroids, alcohol, sickle cell)	Core decompression vs arthroplasty
Bone marrow edema (T2 hyperintense, T1 hypointense), ± sinus tract	Osteomyelitis	Cultures; IV antibiotics 4-6 weeks

Other Modalities

Modality	Key Findings & Interpretation	PANCE Pearls
Mammography (BI-RADS 0-6)	0 = incomplete; 1 = negative; 2 = benign; 3 = probably benign (6 mo f/u, <2% malignancy); 4 = suspicious - biopsy (4A low / 4B intermediate / 4C high); 5 = highly suggestive of malignancy (>95%) - biopsy ; 6 = known biopsy-proven cancer	Suspicious findings: spiculated mass, pleomorphic clustered microcalcifications, architectural distortion, skin/nipple retraction . Density a-d (d = extremely dense → reduced sensitivity)
HIDA scan	GB normally visualizes within 30-60 min. Non-visualization at 1-4 h = acute cholecystitis . EF <35-40% → chronic cholecystitis / biliary dyskinesia	Use when US is equivocal but suspicion remains
VIQ scan	Ventilation/perfusion mismatch (perfusion defect with normal ventilation) = high probability PE	Use when CT-PA contraindicated (renal failure, contrast allergy, pregnancy)
Bone scan	Multiple 'hot spots' = metastases (prostate, breast, lung). Diffuse skull/pelvis uptake = Paget disease . 'Hot nose' on cerebral perfusion scan = brain death sign	Bone pain + known cancer → bone scan first; MRI for cord compression risk
Barium swallow	Bird's beak at LES with proximal dilation → achalasia Apple-core annular narrowing → esophageal/colonic carcinoma Corkscrew esophagus → diffuse esophageal spasm Schatzki ring → intermittent dysphagia for solids	Use water-soluble contrast (Gastrografin) if perforation suspected; barium causes peritonitis
Upper GI / barium enema	String sign of Kantor (terminal ileum) → Crohn disease Lead-pipe colon (haustral loss) → chronic UC Apple-core lesion → colorectal carcinoma	Largely replaced by colonoscopy and CT enterography in practice
DEXA (T-score, WHO criteria)	Normal: T ≥ -1.0 Osteopenia: -1.0 to -2.5 Osteoporosis: ≤ -2.5 Severe (established): ≤ -2.5 + fragility fracture	USPSTF: screen women ≥65 (or younger if elevated FRAX). T-score for postmenopausal women / men ≥50; Z-score otherwise (Z ≤ -2.0 = below expected for age)

Cross-Modality 'Buzzword' Sign Reference

Quick-recognition layer. When a vignette uses one of these phrases, the diagnosis is usually in the right column.

Sign	Modality	Diagnosis
Steeple sign	AP soft tissue neck	Croup
Thumbprint sign	Lateral neck X-ray	Acute epiglottitis
Thumbprinting (mucosal edema)	AXR / CT colon	Ischemic / inflammatory colitis, toxic megacolon
Boot-shaped heart	CXR	Tetralogy of Fallot
Egg on a string	CXR	Transposition of the great arteries
Snowman / figure of 8	CXR	Supracardiac TAPVR
Scimitar sign	CXR	Partial anomalous pulmonary venous return
Figure of 3 / rib notching	CXR	Coarctation of the aorta
Westermarck sign (focal oligemia)	CXR	Pulmonary embolism
Hampton hump (pleural-based wedge)	CXR	Pulmonary embolism (peripheral infarct)
Kerley B lines	CXR	Interstitial pulmonary edema (CHF)
Bat-wing / butterfly opacity	CXR	Alveolar pulmonary edema
Honeycomb lung	HRCT	Idiopathic pulmonary fibrosis (UIP)
Eggshell hilar calcification	CXR	Silicosis
Bulging fissure	CXR	Klebsiella pneumonia
Tree-in-bud	Chest CT	Endobronchial spread (TB, MAC, aspiration)
Ground-glass	Chest CT	PCP, viral pneumonia, alveolar proteinosis
Double bubble	AXR (newborn)	Duodenal atresia (associated with Down syndrome)
Triple bubble	AXR (newborn)	Jejunal atresia
Rigler / double-wall sign	AXR	Pneumoperitoneum
Football sign	AXR (infant supine)	Massive pneumoperitoneum
Coffee-bean / kidney-bean sign	AXR	Sigmoid volvulus
Bird's beak	Barium enema / CT	Sigmoid volvulus / closed-loop obstruction
Bird's beak	Barium swallow	Achalasia
Whirl / whirlpool sign	CT abdomen	Volvulus / mesenteric twist
String sign of Kantor	SBFT / CT enterography	Crohn disease (terminal ileum)
Lead-pipe colon	Barium / CT	Chronic ulcerative colitis
Apple-core lesion	Barium / CT colon	Colorectal carcinoma
Stepladder air-fluid levels	Upright KUB	Small bowel obstruction
Hyperdense MCA	Non-contrast head CT	Acute MCA occlusion (early ischemic stroke)
Loss of gray-white / insular ribbon	Head CT	Early ischemic stroke
Lens-shaped (biconvex) hyperdensity	Head CT	Epidural hematoma (does NOT cross sutures)
Crescentic hyperdensity crossing sutures	Head CT	Subdural hematoma
Star/pentagon hyperdensity in basal cisterns	Head CT	Aneurysmal SAH
Dawson fingers	Brain MRI T2/FLAIR	Multiple sclerosis
Ring-enhancing lesions	CT/MRI brain	Abscess, toxoplasmosis (HIV), GBM, mets
Crescent sign (subchondral lucency)	Hip X-ray / MRI	Avascular necrosis of femoral head

Sign	Modality	Diagnosis
Codman triangle / sunburst periostitis	Bone X-ray	Osteosarcoma
Onion-skin periostitis	Bone X-ray	Ewing sarcoma
Punched-out lytic lesions	Skull X-ray	Multiple myeloma
Salt-and-pepper skull	Skull X-ray	Hyperparathyroidism
Bamboo spine	Spine X-ray	Ankylosing spondylitis
Pencil-in-cup deformity (DIPs)	Hand X-ray	Psoriatic arthritis
Chondrocalcinosis (menisci/cartilage)	Knee X-ray	Pseudogout (CPPD)
Joint-space narrowing + osteophytes + sclerosis + cysts	Joint X-ray	Osteoarthritis
Periarticular erosions, soft-tissue swelling, ulnar deviation	Hand X-ray	Rheumatoid arthritis
Lung sliding / 'seashore' on M-mode	Lung US	Normal pleura (rules OUT pneumothorax at that spot)
Loss of lung sliding / 'barcode' on M-mode	Lung US	Pneumothorax
Lung point	Lung US	Pathognomonic for pneumothorax (transition between sliding and non-sliding)